

Report of the mini-project work done in the related subject w.r.t the habitation/village.

An IoT Based Air Quality Monitoring System in which we will monitor the Air Quality and will trigger a alarm when the air quality goes down beyond a certain level, means when there are sufficient amount of harmful gases are present in the air like CO₂, smoke, alcohol, benzene and NH₃. It will show the air quality in PPM on the so that we can monitor it very easily.

we have used MQ135 sensor as the air quality sensor which is the best choice for monitoring Air Quality as it can detects most harmful gases and can measure their amount accurately.

First of all we will connect the ESP8266 Node Mcu. ESP8266 runs on 3.3V and if you will give it 5V then it won't work properly and it may get damage. 24 ESP8266 Wi-Fi module gives your projects access to Wi-Fi or internet. It is a very cheap device and make your projects very powerful. It can communicate with any microcontroller and it is the most leading devices in the IOT platform. Learn more about using ESP8266 with Arduino [here](#).

Then we will connect the MQ135 sensor with the Node Mcu. Connect the VCC and the ground pin of the sensor to the Vin and ground of the Node Mcu and the Analog pin of sensor to the A0 of the Node Mcu.

CONNECTIONS:

1. Mq135 sensor's A0 pin is connected to A0 pin of NodeMcu.
2. Mq135 sensor's ground pin is connected to NodeMcu's ground pin.
3. Mq135 sensor's Vcc pin is connected to Vin of NodeMcu.
4. LCD display is connected to I2C module.
5. I2C module's Vcc pin is connected to Vin of NodeMcu
6. I2C module's Ground pin is connected to ground pin of NodeMcu.
7. I2C module's SDA pin and SCL pins are connected to D2 and D1 pins of NodeMcu respectively.
8. Three LED's are connected to the D5, D6, D7 pins of NodeMcu respectively along with resistors.

9. Buzzer is connected to D8 pin of NodeMcu.
10. Some resistors are connected for LED's and buzzer with respect to ground.

The MQ135 sensor can sense NH₃, NO_x, alcohol, Benzene, smoke, CO₂ and some other gases, so it is perfect gas sensor for our Air Quality Monitoring Project. When we will connect it to NodeMcu then it will sense the gases, and we will get the Pollution level in PPM (parts per million). MQ135 gas sensor gives 25 the output in form of voltage levels and we need to convert it into PPM. So for converting the output in PPM, here we have used a library for MQ135 sensor, it is explained in detail in “Code Explanation” section below.

Sensor was giving us value of 90 when there was no gas near it and the safe level of air quality is 350 PPM and it should not exceed 400 PPM. When it exceeds the limit of 400 PPM, then it starts cause Headaches, sleepiness and stagnant, stale, stuffy air and if exceeds beyond 1000 PPM then it can cause increased heart rate and many other diseases.

When the value will be less than 350 PPM, then the LCD will display “Fresh Air”. Whenever the value will increase 400 PPM, then the buzzer will start beeping and the LCD and webpage will display “Danger”. If it will increase 800 then the buzzer will keep beeping and the LCD and webpage will display “Danger! Move to fresh Air”.

OBJECTIVES:

To combine advanced detection technologies to produce an air quality sensing system with advanced capabilities to provide Low-cost comprehensive monitoring. To display the sensed data in user friendly format in LCD display panel.

HARDWARE REQUIREMENTS:

1. NodeMcu.



2. MQ135 gas sensor.



3. LCD display (16x2)



4. I2C module



5. Buzzer.



6. Leds (Red, Blue, Green)



7. Resistors



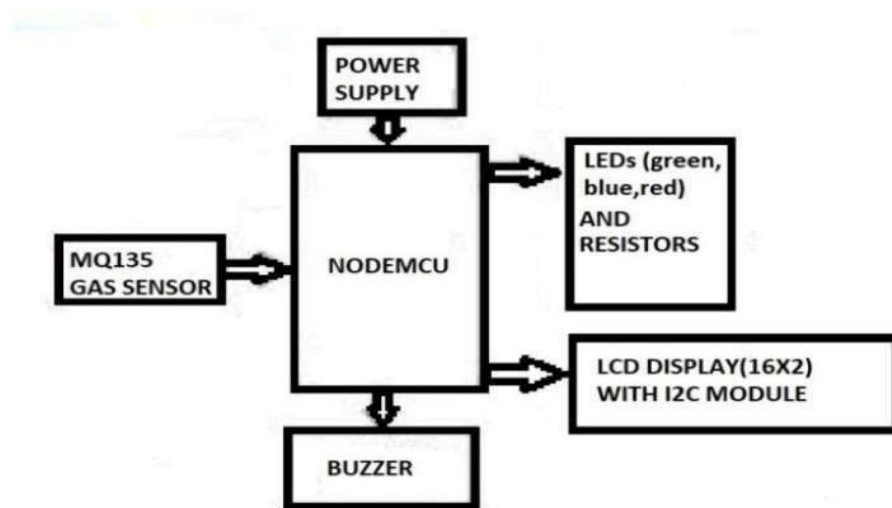
8. some jumping wires



9. Bread board

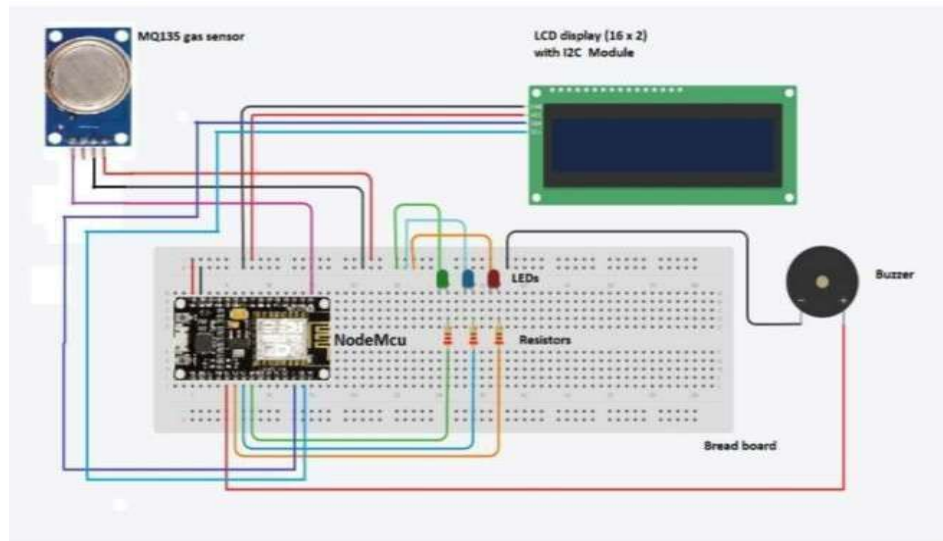


SYSTEM DESIGN (BLOCK DIAGRAM):



IMPLEMENTATION:

Circuit Diagram



Code modulus

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16,2);
int led1=D5;
int led2=D6;
int led3=D7;
int buzzer=D8;
int mqsensor=A0;
void setup() {
  Serial.begin(9600);
  pinMode(led1,OUTPUT);
  pinMode(led2,OUTPUT);
  pinMode(led3,OUTPUT);
  pinMode(buzzer,OUTPUT);
  pinMode(mqsensor,INPUT);
  lcd.init();
  lcd.clear();
  lcd.backlight();
  lcd.setDelay(0,0);
  lcd.print("hello");
}
void loop(){ // int sensor=100;
  int sensor=analogRead(mqsensor);
  if(sensor<=350)
```

```

{
digitalWrite(led1,1);
digitalWrite(led2,0);
digitalWrite(led3,0);
Serial.println(sensor);
lcd.setCursor(0,0);
lcd.print("AirPollution:");
lcd.setCursor(13,0);
lcd.print(sensor);
lcd.setCursor(0,1);
digitalWrite(buzzer,0);
lcd.print("Condition: fresh");
}
else if(sensor<=400 && sensor>350)
{
digitalWrite(led3,0);
digitalWrite(led1,0);
digitalWrite(led2,1);
Serial.println(sensor);
lcd.setCursor(0,0);
lcd.print("AirPollutio:");
lcd.setCursor(13,0);
lcd.print(sensor);
lcd.setCursor(0,1);
digitalWrite(buzzer,0);
lcd.print("Condition:average");
}
else if(sensor<=600 && sensor>400)
{
digitalWrite(led3,1);
digitalWrite(led1,0);
digitalWrite(led2,0);
Serial.println(sensor);
lcd.setCursor(0,0);
lcd.print("AirPollutio:");
lcd.setCursor(13,0);
lcd.print(sensor);
lcd.setCursor(0,1);
digitalWrite(buzzer,1);
lcd.print("Condition:danger");
}
delay(1000);
}

```

EXPERIMENTAL RESULTS:

The air quality monitoring system device has given the output as

AirPollution: 328

Condition: fresh

The result of the experiment done in the place Yanamalakuduru has been shown in the below photos section.

CONCLUSION AND LIMITATIONS CONCLUSION:

Conclusion

This research proposed a smart Air Quality monitoring system that constantly keeps track of air quality in an area and displays the air quality measured on an LCD screen. It turns on a green light when the atmosphere is fresh. Otherwise it turns on a blue light when some considerable amount of pollutants are present in atmosphere. If the pollution level is high then the device turns on red light which indicates that it is danger.

Limitations:

The device lacks a calibration technique. It shows readings that are a bit more or less than the actual readings.

PHOTOS

